Electronics 2: Detectors

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How to think of detection

A way to convert physical phenomena into measurable signals.

A measurand is the quantity we are trying to measure.

Transducers are devices that convert measurements into electrical signals.

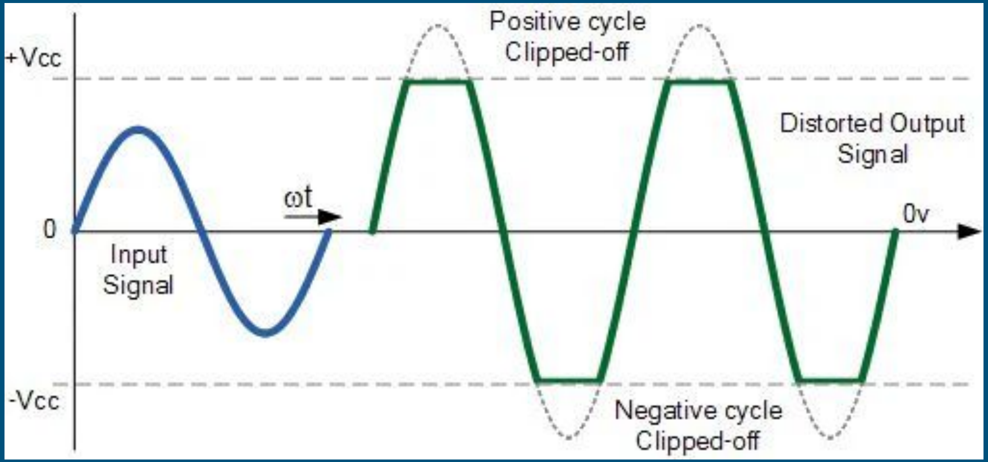
Consideration in Detection

When picking a detector, we need to ask:

1. **Dynamic range:** As the signal changes strength can I always measure it?
2. **Responsivity**: How strong will the signal produced by our detector be?
3. **Coupling:** How efficiently can the measurement be coupled to the detector?
4. **Noise**: What effects will obscure the signal we're measuring?
5. **Backaction**: How will our detector itself affect our measurement?
6. **Detection method:** Will we be if something changes at all? Measuring changes from 0? A difference between two values?

Dynamic Range

Dynamic range is a detector's ability to capture a signal over its full range of power.

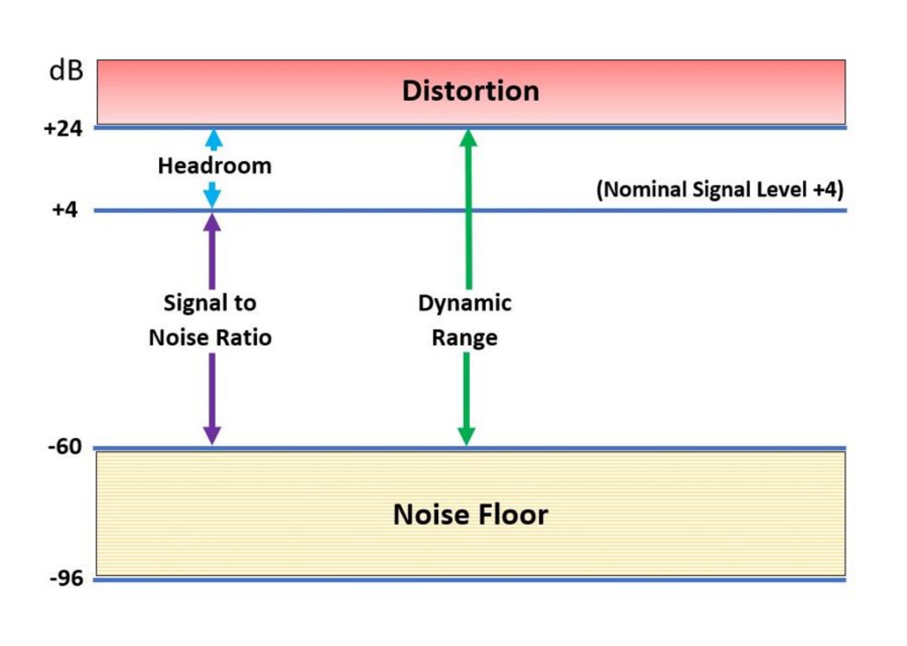


Dynamic Range

Dynamic range is a detector's ability to capture a signal over its full range of power.

Our noise floor plays an important part of determining the range we can accurately capture our signal.

More noise > smaller margin before distortion



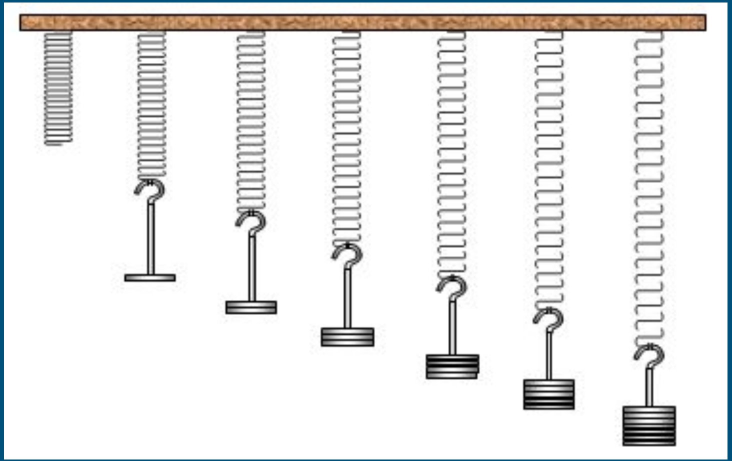
Responsivity

We want to measure a quantity by converting the signal into something easier to observe.

With a mass on a spring, the length of the spring is the observable, our responsivity is strongly related to the spring constant.

Imagine k is low. How would that impact our responsivity? What would be the benefits or concerns?

What if k was high?



$$F = kx$$

F = Force applied on the spring
k = spring constant
x = extention of the spring

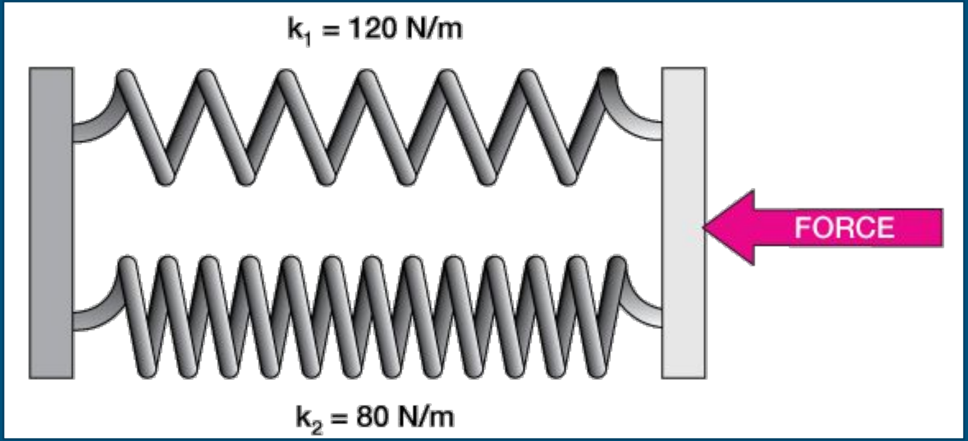
Responsivity

High K - very stiff spring

- Will be able to measure up to higher masses
- Less sensitive to fluctuations
- δx will be significantly smaller, less accuracy.

Low K - loose spring

- Can't measure high masses
- More sensitive to oscillations
- δx will be significantly larger, easier to measure.

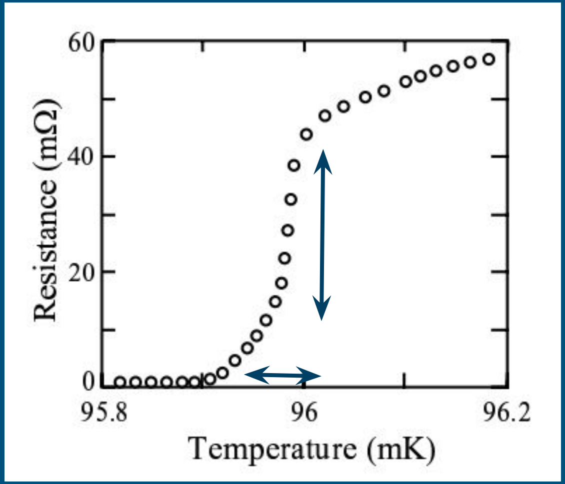


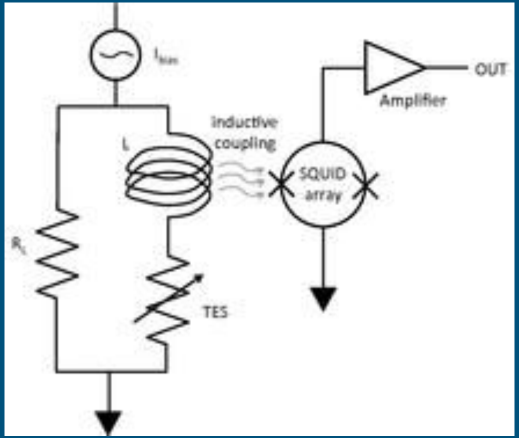
Responsivity - TES

With transition edge sensors, we leverage the property of superconductors:

Within the superconducting transition, small changes in T -> Large changes in R.

We can then convert this to an electric signal by coupling an inductor with changing current to a SQUID, which detects change in magnetic flux.

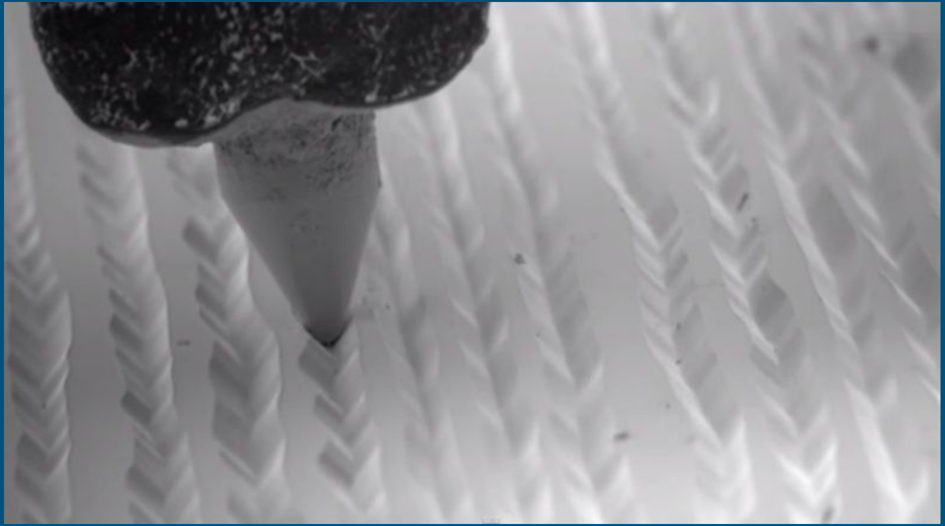




Back Action: Every action has an Equal Reaction

Any measurement will affect the physical phenomenon to some level. Ie. Newton’s Third law.

This can be called **backaction**, where our measurement **itself** will disturb the system and affect the results.

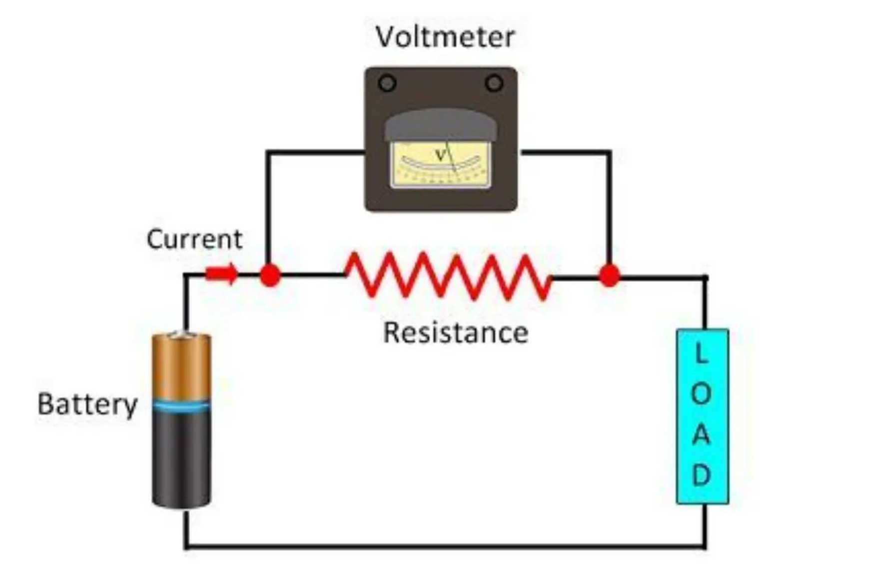


Backaction Example - Voltmeter

Imagine measuring the voltage across a resistor.

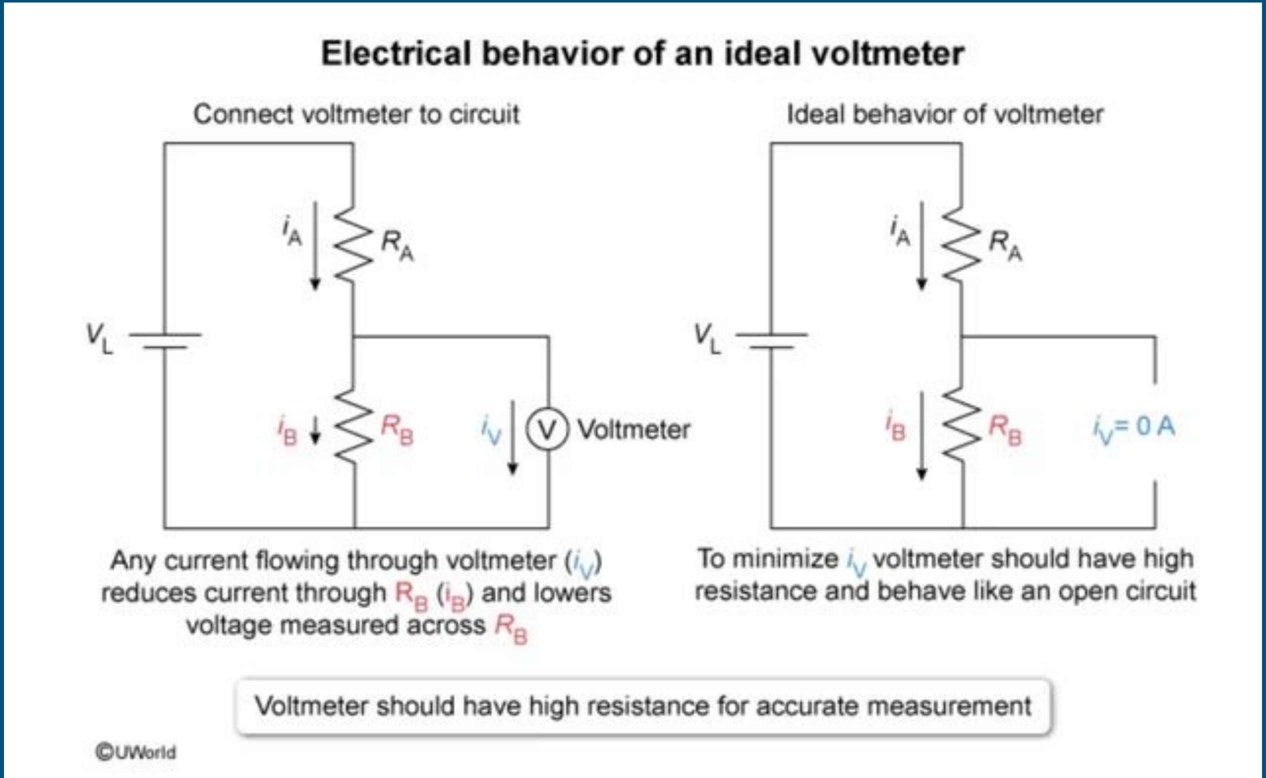
The voltmeter will be attached in **parallel** with the resistor.

$$R_{tot} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$



The voltmeter being introduced to the circuit will change the voltage across the resistor. How can we minimize the impact of the voltmeter?

Backaction Example - Voltmeter



Measurement Types

Absolute measurements

Measuring values like fundamental constants.

Ex. Mass of neutrino, charge of electron, speed of light

Require high ‘accuracy’

Relative measurements

The difference between two measurements using the same system.

Ex. Mixing angle of 2 neutrino flavours, pH, redshift of a galaxy, CMB anisotropies

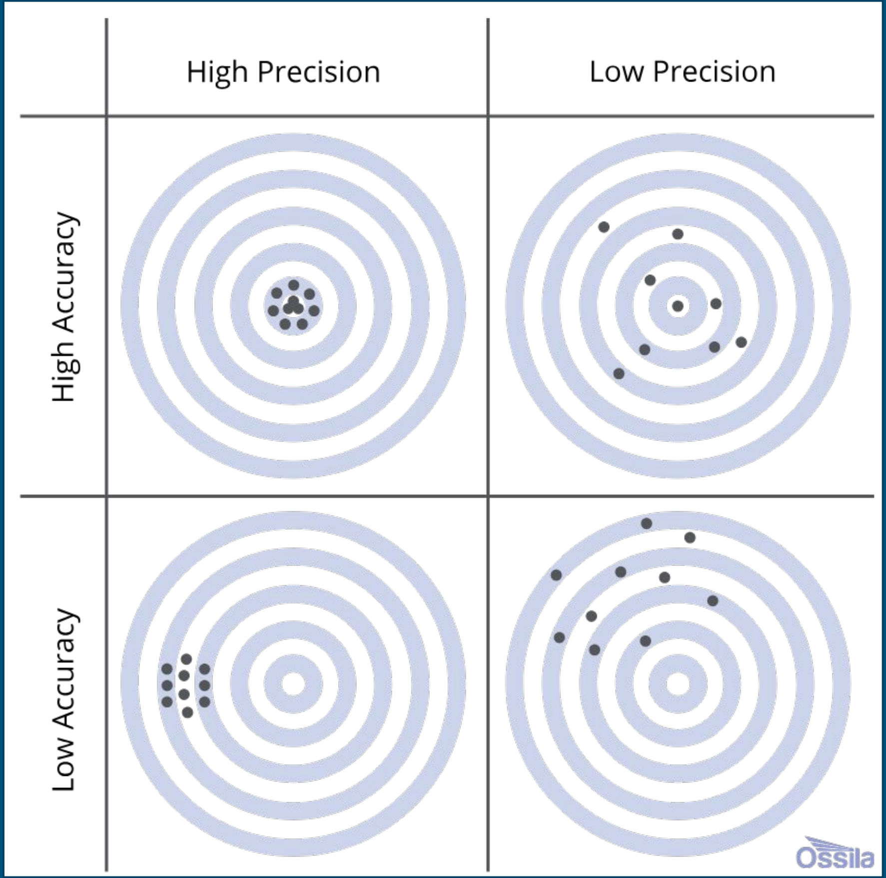
Requires high ‘precision’

Which type of measurement do you think is harder?

Precision vs. Accuracy

To do an absolute measurement requires **both** high accuracy and precision.

To do a relative measurement only really requires high accuracy.

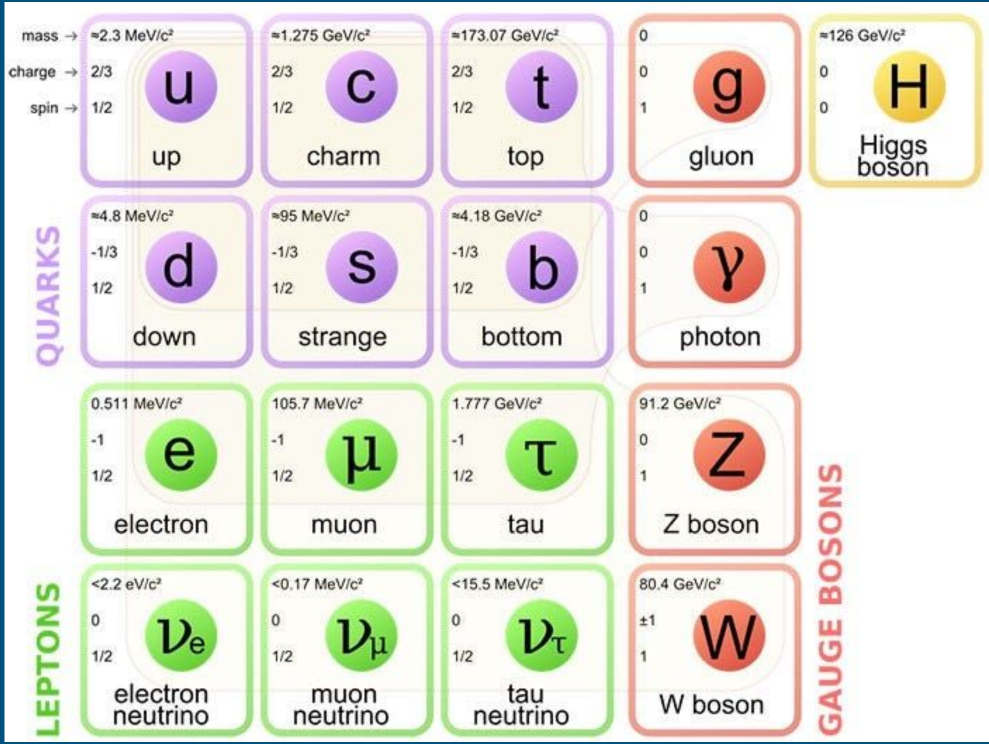


Absolute Measurement: Muon G-2

In particle physics, the Standard Model predicts with high accuracy many fundamental quantities - charge, spin, mass… So far has been remarkably correct.

A question emerges:

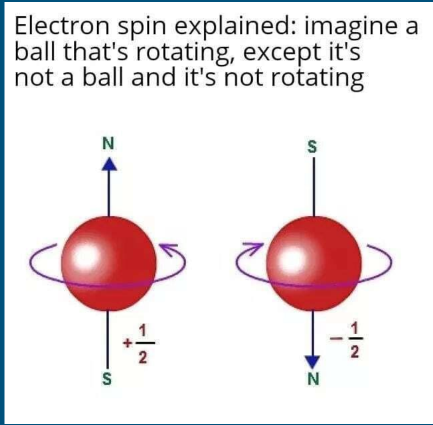
Can new physics be discovered if we measure precisely determined particle quantities to deviated from theory?

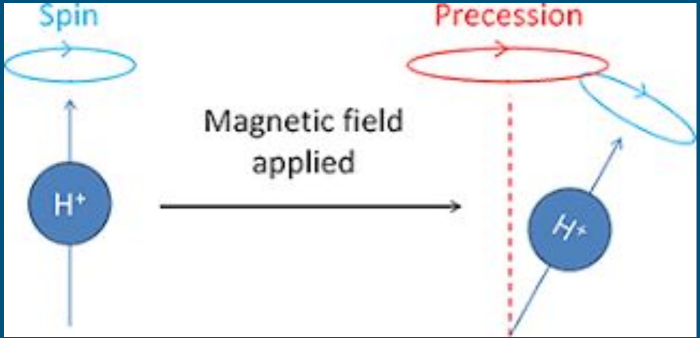


Absolute Measurement: Muon G-2

A particle has intrinsic angular momentum, which we call ‘spin’.

When a magnetic field is applied, the spin axis will precess. The ratio of a particle’s spin to its magnetic moment is called ‘g’, the gyromagnetic ratio.



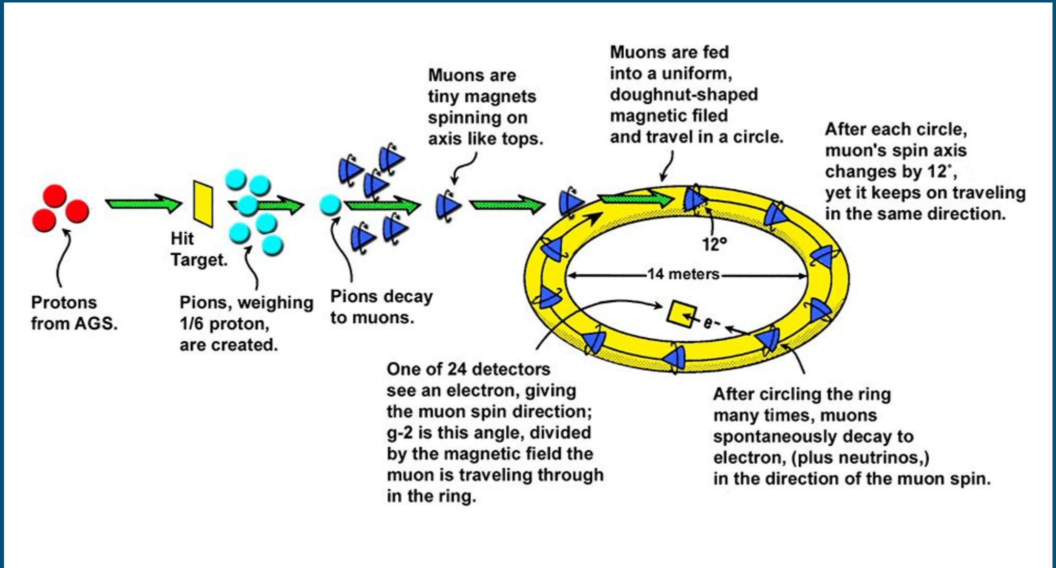


Absolute Measurement: Muon G-2

Muon G-2 is an experiment at Fermilab that is measuring any anomalies in the g of a muon.

Shoots 3 GeV muons into an **extremely** uniform magnetic field.

As muons decay, higher energy electrons will be preferentially emitted in spin direction and detected.



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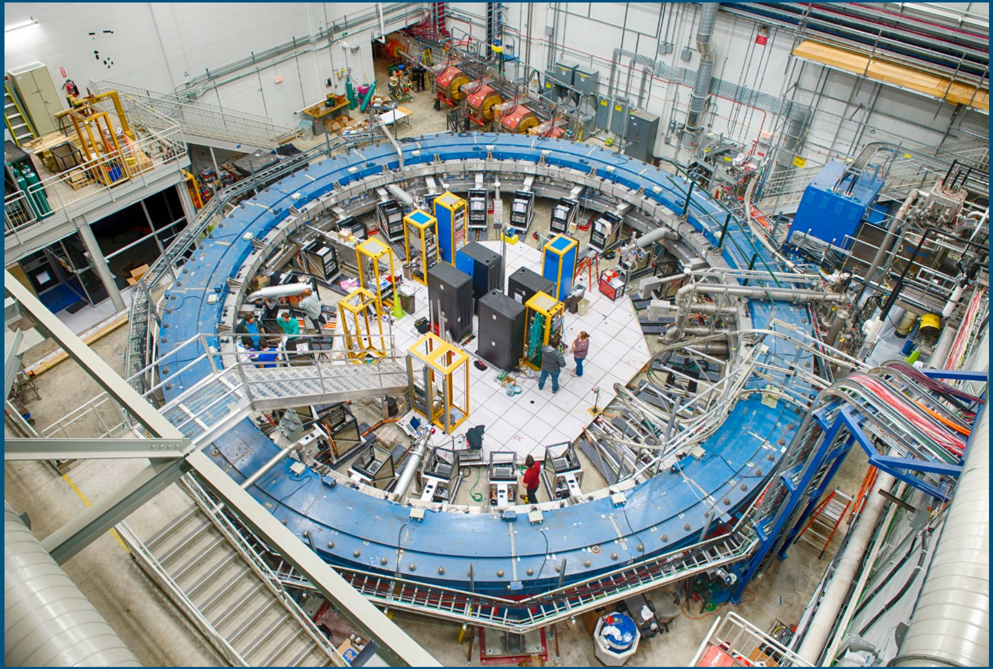
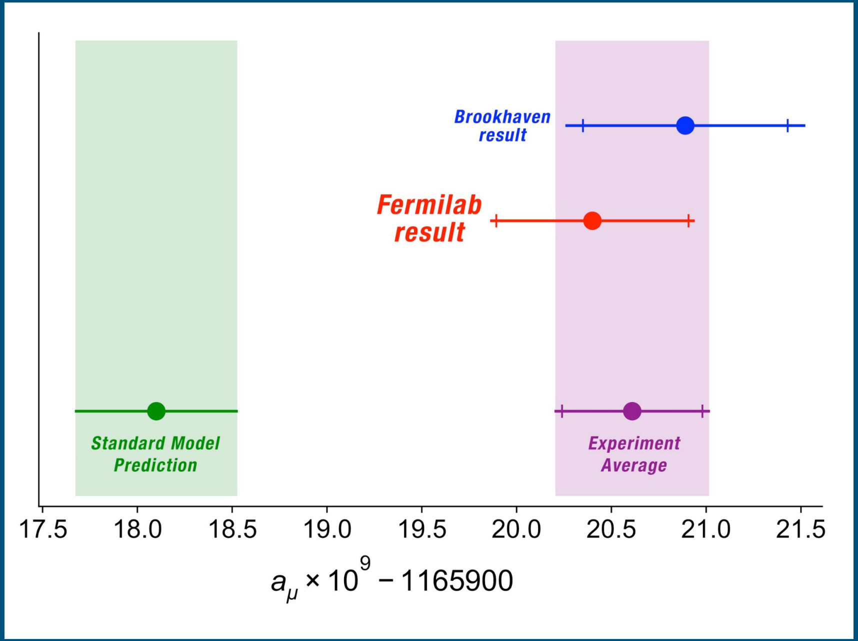


Absolute Measurement: Muon G-2

G-2 successfully measured a deviation at the **parts per billion level** of a_μ, the anomalous magnetic moment of a muon.

Required the analysis of > 8 billion individual muons!

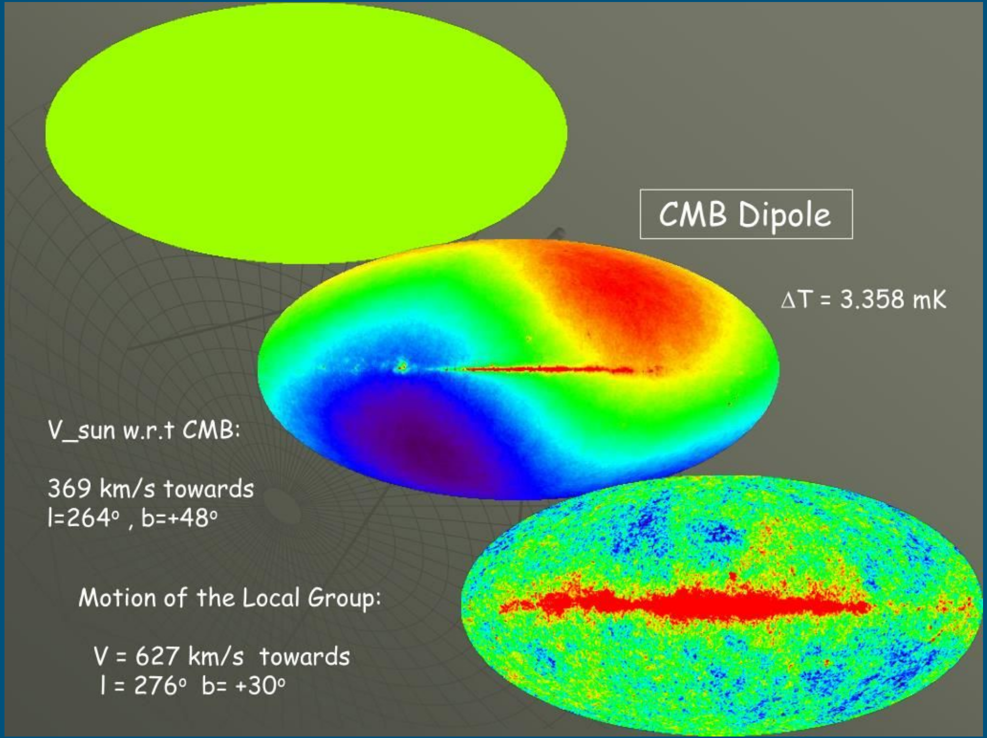


$$a_\mu = \frac{g-2}{2}$$

Relative Measurement: CMB Anisotropies

The Cosmic Microwave Background is the leftover light from the big bang, and still fills our universe to this day.

It has been measured in all directions at 2.73K but mean subtraction reveals more structure…

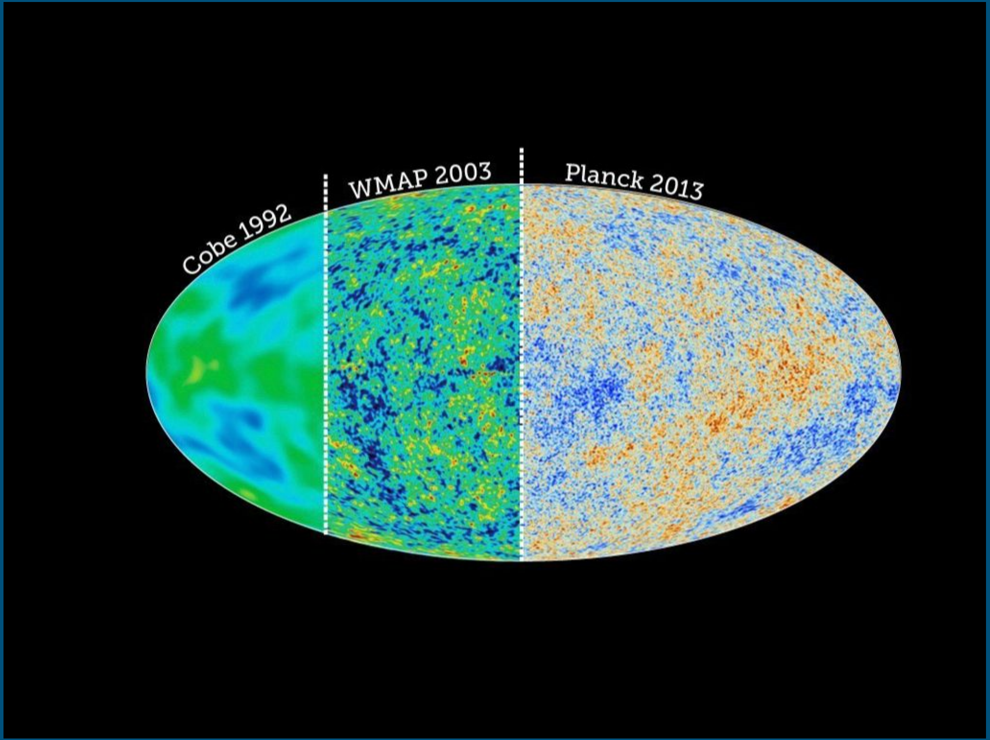


Relative Measurement: CMB Anisotropies

COBE was the first experiment to measure temperature anisotropies in the CMB.

Overtime our relative measurement of the hot and cold spots of the CMB has become finer and finer!

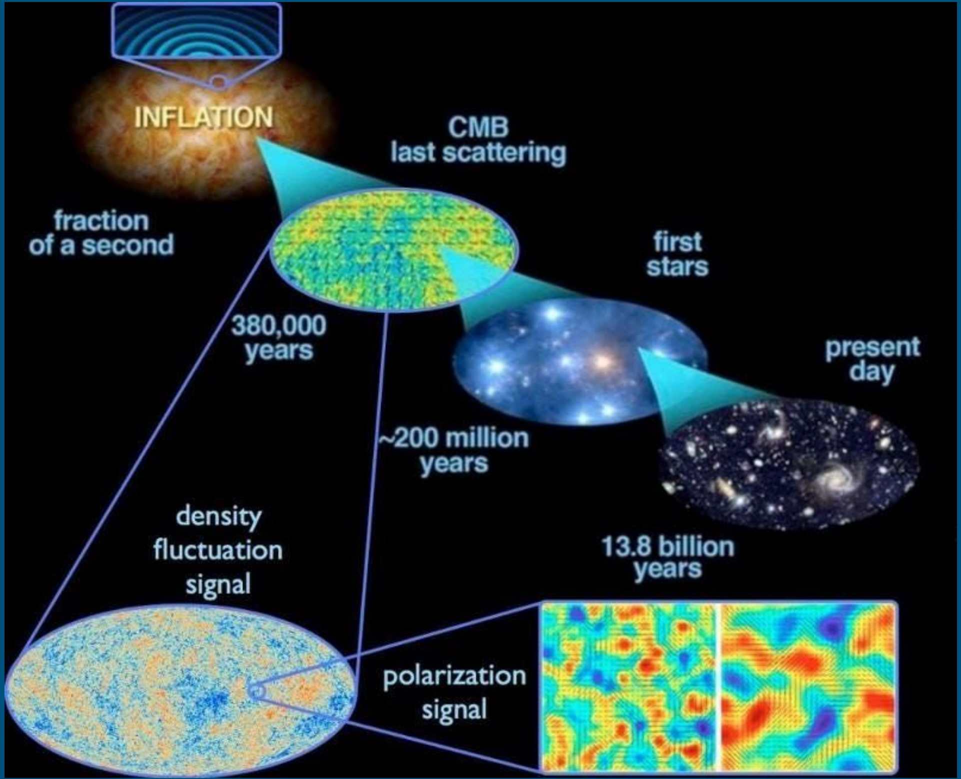
What are the origin of these fluctuations?



Relative Measurement: CMB Anisotropies

We believe that quantum fluctuations from dense early universe expanded suddenly and got ‘frozen’ into the universe.

These hot/cold spots then seed the initial conditions for matter to collect to form the first stars -> galaxies today!



Absolute vs. Relative Measurements

Absolute measurements generally are of fundamental physical constants, or measuring the existence/nonexistence of something. Ie. Speed of light, higgs boson.

If your system is very noisy, or things can mimic the signal you’re trying to measure, it will be really difficult to convince yourself you’re seeing the right thing!

Relative measurements are leveraging the fact that you are measuring your signal as the change in response on your system. Therefore you are comparing the **change in signal** to the **change in noise.**

Measurement Methods

Approximately can categorize as:

Deflection method: Result directly related to measurand. Output depends on readout.

Difference method: Measuring change of measurand to a reference.

Null method: Measuring that something does **not** occur. Depends solely on properties of reference, and accuracy of the null obtained.

Deflection Method

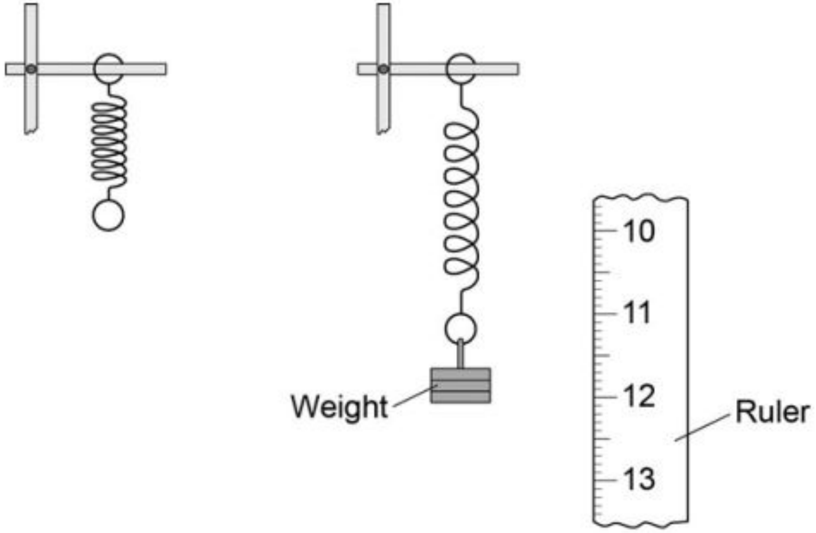
Say we place a weight on a spring. dX will be directly related to M, the weight.

The output is determined by Hooke's law of the spring, our 'detector':

$$\boxed{F = kx}$$

F = Force applied on the spring
k = spring constant
x = extention of the spring

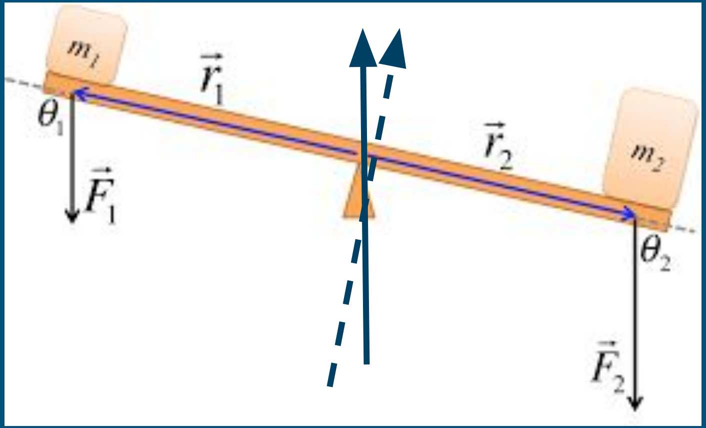
Figure 1



Difference Method

When we compare the weight of two objects, we are using the same system to measure both - we are only measuring the difference between a reference and the measurand.

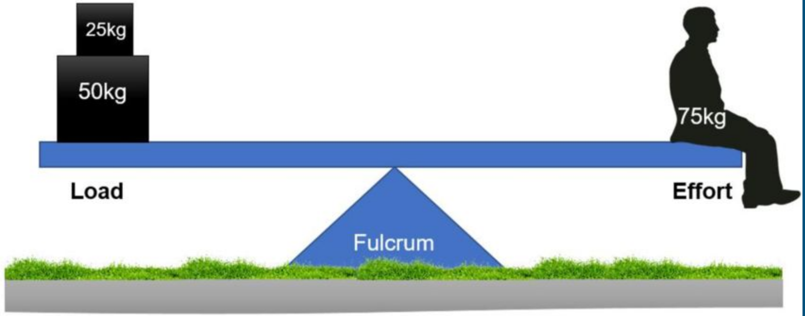
Our output will therefore depend both on our measurement of the reference and of the system.



Null Method

Similar to measuring a difference, but now we want to be able to tell when two quantities are **exactly equal.** Ie. the seesaw is exactly flat.

Will depend on the accuracy of our setup being very high, and us trusting well our reference.



When to use which method?

Low background, signal can be fully measured by detector: Deflection method

Large background, small signal all within dynamic range: Differencing or null

But what if we have a small signal, but the background is so strong it exceeds the dynamic range of our system? We will need to **amplify** the signal we want to measure to be higher than our background noise.